LLD — iClinic AI Front Desk Assistant

Low Level Design: AI Front Desk Assistant (Healthcare)

Parent Epic: iClinic — AI Front Desk Assistant
Module: Appointment Booking via Voice & AI Outbound Calls
Status: In Design
Phase: Phase 1 (Voice + Outbound) | Phase 2 (Chat)
Last Updated: 2026-02-25

Problem Statement

Clinics often rely on manual front desk staff to handle appointment scheduling through phone calls or in-person visits. This process leads to long wait times, limited availability outside working hours, booking errors, missed appointments, and increased workload for staff.

The goal is to design and implement an **AI-powered front desk assistant for iClinic** that allows patients to schedule, modify, and confirm doctor appointments through natural **voice conversations, AI outbound phone calls, and chat interfaces**, while integrating with the clinic's existing scheduling system.

User Story

As a patient, I want to book an appointment with my preferred doctor using a natural voice conversation or phone call, so that I receive instant confirmation without visiting or calling the front desk manually.

Business Impact

- Patient Experience: Faster booking with 24/7 availability via voice and outbound calls
- Front Desk Workload: Reduced routine scheduling calls handled by the AI assistant
- Slot Utilization: Alternate slot suggestions reduce no-slot-found abandonment
- Confirmation Accuracy: Automated SMS/Email reduces missed appointments
- Scalability: Outbound call campaigns enable proactive scheduling at scale

Functional Requirements

Requirements Summary

The iClinic AI Front Desk Assistant shall enable end-to-end conversational appointment booking through Voice Assistant, AI Outbound Calls, and Chat Interface. The system shall integrate with the clinic's existing scheduling system or manual slot feed and manage the complete booking lifecycle — from greeting to confirmation and notification delivery.

Booking Channels

The system shall support appointment booking through the following channels:

- Voice Assistant
- AI Outbound Calls
- Chat Interface

Each channel must provide a consistent booking experience and interact with the same scheduling backend.

Conversational Booking Flow

The assistant shall manage a structured yet natural booking conversation that includes:

1. Greeting the patient
2. Capturing appointment intent or reason
3. Identifying appointment type
4. Asking provider preference (if applicable)
5. Suggesting available slots
6. Confirming selected date and time
7. Reconfirming appointment details
8. Finalizing booking
9. Sending confirmation

The system must support natural conversational behavior, including:

- Interruptions
- Mid-flow corrections (date, time, provider, appointment type)
- Context retention without restarting the flow

Slot Management and Alternate Suggestions

Slot availability shall be retrieved from the clinic's scheduling input (manual feed or integrated system).

The assistant shall:

- Suggest available slots based on user preference
- Offer nearest available alternatives if the requested slot is unavailable
- Suggest same-provider alternate timings
- Suggest next available provider (if patient permits)

Before final booking, the assistant shall reconfirm:

- Date and time
- Provider (if applicable)
- Location (if applicable)
- Appointment type

AI Outbound Calling Requirements

The system shall be capable of initiating outbound calls for appointment scheduling.

During outbound calls, the assistant shall:

1. Initiate the call

2. Perform basic patient identity confirmation (as per clinic policy)
3. Conduct conversational booking
4. Confirm appointment details
5. Send confirmation notification

Booking Confirmation and Notifications

After successful booking, the assistant shall:

• Confirm appointment details within the active conversation
• Send confirmation via SMS and/or Email

Confirmation messages must include:

• Date and time
• Provider (if assigned)
• Location
• Appointment type
• Any applicable instructions

Chat-Based Context-Aware Booking

The chat interface shall operate using conversation state and user context.

The assistant shall capture, at minimum:

• Appointment reason
• Appointment type
• Preferred date and time
• Provider preference (if applicable)
• Basic patient identity confirmation (as per clinic policy)

The chat flow shall guide the user through booking while preserving conversation continuity and handling corrections seamlessly.

Non Functional Requirements

Reliability: The system targets 99.9% uptime, enforced through health checks, auto-restart policies, circuit breakers, and Kubernetes liveness probes.

Security: All data in transit is encrypted with TLS 1.3. Data at rest uses AES-256 encryption in line with HIPAA-aligned storage policies.

Observability: Full stack monitoring is implemented via CloudWatch logs, X-Ray traces, and a custom booking metrics dashboard.

Compliance: PII is handled per HIPAA requirements. Data retention policies are enforced at the database level.

Scalability: The system is designed to support 50 or more concurrent voice sessions using horizontal pod autoscaling. The telephony platform handles call-level concurrency.

Maintainability: The design uses service layer separation, dependency injection, and modular LangGraph flow definitions to support long-term maintainability.

Accessibility: Natural-sounding TTS voices are used throughout. The assistant supports interruptions and corrections at all flow steps.

Use Cases and Success Criteria

UC ID	Use Case	Success Criteria
UC-01	Voice booking, preferred slot available	Patient completes booking end-to-end and receives SMS/Email confirmation.
UC-02	Voice booking, preferred slot unavailable	Assistant offers alternate slots; patient selects one and booking completes.
UC-03	Voice booking, mid-flow correction	Patient corrects date, time, or provider; assistant adjusts without restarting the flow.
UC-04	AI Outbound Call, end-to-end booking	System initiates call, verifies identity, books appointment, and sends confirmation.
UC-05	Chat booking, end-to-end	Patient books via chat with the same capability as voice/call.
UC-06	Chat, alternate slot and correction	Chat handles alternate suggestion and mid-flow corrections seamlessly.

User Journeys and Flows

Voice Booking Flow

1. Greeting — assistant introduces itself and the clinic
2. Capture patient intent — determine if booking, reschedule, or inquiry
3. Identify appointment type — based on reason provided
4. Ask provider preference — preferred doctor or any available
5. Suggest available slots — from the clinic scheduling system
6. Confirm date/time — patient selects or requests alternate
7. Reconfirm booking details — date, time, provider, type, and location
8. Confirm booking — write record to the appointments system
9. Send confirmation — SMS and/or Email dispatched to patient

AI Outbound Call Flow

1. AI initiates call to patient from the outbound_call_tasks queue
2. Identify patient — verify basic identity per clinic policy
3. State purpose — inform patient about the scheduling need
4. Offer available slots — suggest times from available_slots
5. Book appointment — patient confirms; write appointment record to DB
6. Send confirmation — SMS and/or Email dispatched to patient

Chat Booking Flow

1. Greeting
2. Ask reason
3. Suggest appointment types
4. Ask provider preference
5. Suggest available slots
6. Confirm date & time
7. Reconfirm full details
8. Create appointment record
9. Send confirmation (SMS / Email)
10. Provide final confirmation

In Scope

- Voice Assistant appointment booking, AI Outbound Call scheduling, slot suggestion with alternate slot recommendation, natural conversation handling (interruptions and corrections), in-conversation booking confirmation, and SMS/Email confirmation with appointment instructions and basic patient identity confirmation.
- Chat-based appointment booking.
- Calendar invite attachment and clinic staff notification.

Out of Scope

- Provider schedule setup, working hours, breaks, and time-off management (handled manually by front desk)
- Appointment slot generation and blocking (managed via manual/existing clinic system)
- Payment processing or insurance verification
- UI/UX changes outside this story's scope
- Performance optimization beyond baseline requirements

Assumptions & Constraints

Technical: The backend is built on FastAPI with async SQLAlchemy 2.0. The frontend uses React 19. LangGraph orchestrates the conversational flow in a background worker, call processing.

Business: Slot availability is pre-populated manually by the front desk into the available_slots table. Clinic appointment types and durations are pre-configured before the assistant goes live. Client rules are pre-configured before processing.

Data: PostgreSQL is the primary database with proper indexing across all tables.

Integration: Twilio or an equivalent telephony platform is used for voice and outbound calls.

Legal: All patient data handling complies with HIPAA and applicable privacy regulations.

Identity: Patient identity confirmation policy is defined by the clinic and is configurable per deployment.

API/Contracts

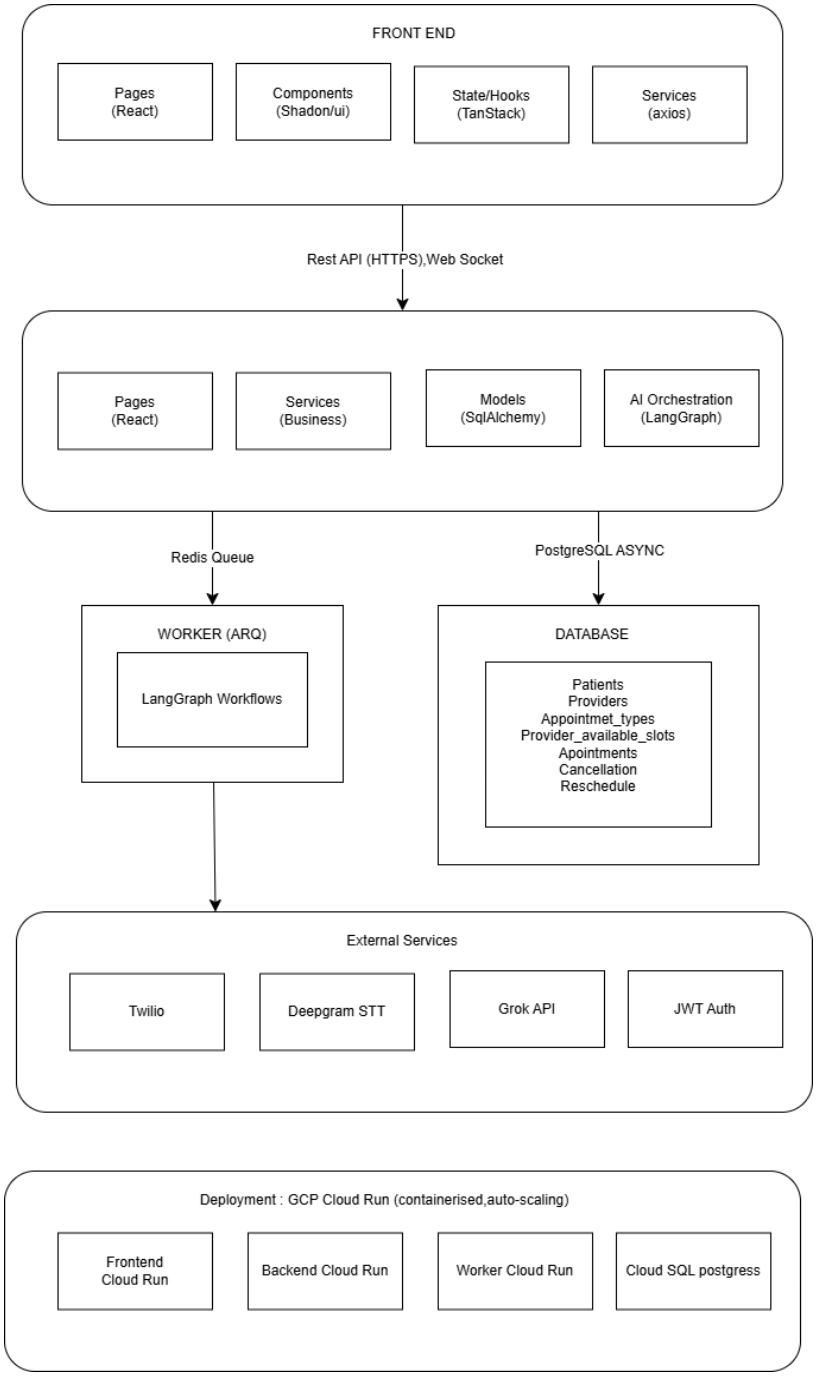
Internal APIs

Method	Endpoint	Description	Auth
GET	/api/v1/appointments	List appointments with filtering by patient, provider, date, and status	JWT Required
POST	/api/v1/appointments	Create a new appointment (booking)	JWT Required
PUT	/api/v1/appointments/{id}	Update an appointment (reschedule or cancel)	JWT Required
GET	/api/v1/available-slots	Query available slots by provider, date, and appointment type	JWT Required
GET	/api/v1/providers	List active providers with specializations	JWT Required
GET	/api/v1/appointment-types	List active appointment types and durations	JWT Required
POST	/api/v1/outbound-tasks	Create an outbound call task for a patient	JWT Required
GET	/api/v1/outbound-tasks/{id}	Get outbound call task status and attempts	JWT Required
POST	/api/v1/conversations/sessions	Initialize a new conversation session	JWT Required
PUT	/api/v1/conversations/sessions/{id}	Update session state (context, status)	JWT Required

Method	Endpoint	Description	Auth
POST	/api/v1/notifications	Dispatch an SMS or Email confirmation	JWT Required

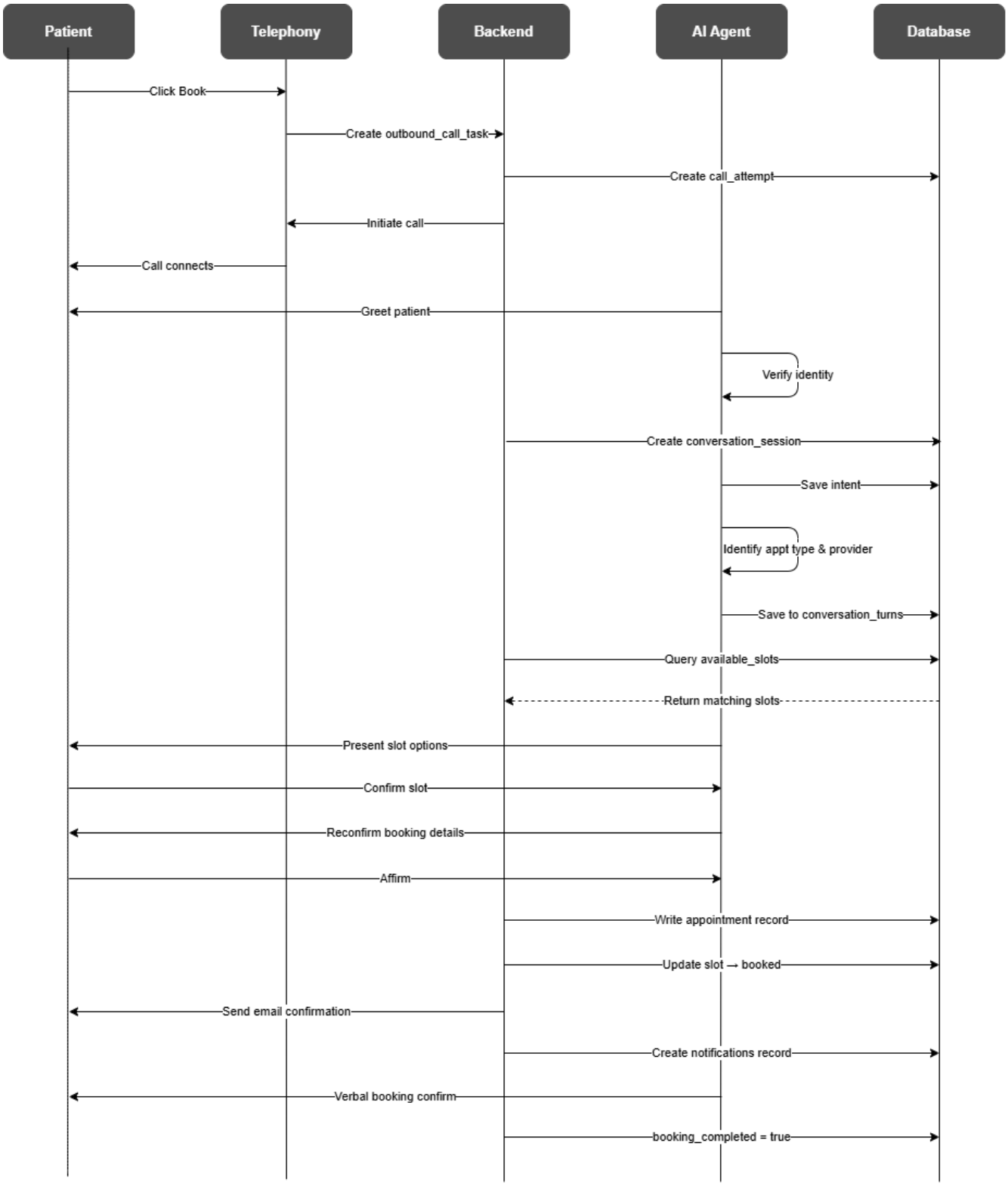
Design

High Level Architecture



Low Level Design

Sequence Diagram



Technical Assumptions

- Async/await patterns are used throughout the FastAPI backend
- Database transactions are managed via SQLAlchemy async sessions
- All APIs follow RESTful conventions with versioned paths (/api/v1/)
- Authentication uses JWT Bearer tokens
- LangGraph orchestrates the multi-turn conversation state machine
- The telephony platform handles call initiation, audio streaming, and voice input
- Available slots are managed externally by the front desk; the assistant reads but does not write slot generation
- Conversation context is stored as JSONB in conversation_sessions.context_snapshot for LangGraph state recovery

Approach Analysis

- Telephony provider selection (Twilio vs. alternatives) and per-minute cost model
- Async/await patterns used throughout backend
- Database transactions managed via SQLAlchemy session
- API follows RESTful conventions
- Authentication via JWT tokens

Open Questions

- Telephony provider selection (Twilio vs. alternatives) and per-minute cost model
- Voice AI model selection (ElevenLabs, Google TTS, Twilio Voice) for natural-sounding responses
- Patient identity verification level — date of birth, name, phone match, or PIN — per clinic policy
- Max concurrent outbound call concurrency and rate-limiting rules
- Error retry policies fine-tuning for failed call attempts
- Performance benchmarks for production voice call load

Request / Response Schema

POST /api/v1/appointments — Request Example

```
{
  "patient_id": 42,
  "provider_id": 5,
  "appointment_type_id": 2,
  "availability_slot_id": 101,
  "session_id": 88,
  "reason_for_visit": "Annual checkup",
  "booking_channel": "voice"
}
```

POST /api/v1/appointments — Response Example

```
{
  "id": 220,
  "status": "scheduled",
  "scheduled_date": "2026-03-10",
  "scheduled_start_time": "10:00:00",
  "scheduled_end_time": "10:30:00",
  "provider": "Dr. Sarah Lee",
  "created_at": "2026-02-25T09:15:00Z",
  "message": "Appointment booked successfully"
}
```

Authentication / Authorization

- Current: JWT Bearer token in the Authorization header
- Roles: admin, front_desk, patient, provider, viewer

Data Models

Database Tables

The following tables represent the full data model for the iClinic AI Front Desk Assistant. Only these tables are represented as structured tables in this document. PK = Primary Key, FK = Foreign Key, Y = Yes.

Users

Field Name	Data Type	Constraint	PK	FK	Description
id	INTEGER	PRIMARY KEY	**Y**		Auto-increment unique user ID
role_id	INTEGER	NOT NULL		**Y**	FK → roles.id
first_name	VARCHAR(100)	NOT NULL			User first name
last_name	VARCHAR(100)	NOT NULL			User last name
email	VARCHAR(255)	UNIQUE NOT NULL			Login email address
phone_no	VARCHAR(20)	NULLABLE			Contact phone number
password	VARCHAR(255)	NOT NULL			Hashed password (bcrypt)
created_at	TIMESTAMP	DEFAULT NOW()			Record creation timestamp

Roles

Field Name	Data Type	Constraint	PK	FK	Description
id	INTEGER	PRIMARY KEY	**Y**		Auto-increment role ID
role_name	VARCHAR(50)	UNIQUE NOT NULL			Role name e.g. admin, front_desk, patient
created_at	TIMESTAMP	DEFAULT NOW()			Record creation timestamp
updated_at	TIMESTAMP	DEFAULT NOW()			Last updated timestamp

Permissions

Field Name	Data Type	Constraint	PK	FK	Description
id	INTEGER	PRIMARY KEY	**Y**		Auto-increment permission ID
name	VARCHAR(100)	UNIQUE NOT NULL			Permission identifier e.g. appointments:create
description	VARCHAR(255)	NULLABLE			Human-readable description
module	VARCHAR(100)	NOT NULL			Module owning this permission
is_active	BOOLEAN	DEFAULT TRUE			Whether permission is active
created_at	TIMESTAMP	DEFAULT NOW()			Record creation timestamp
updated_at	TIMESTAMP	DEFAULT NOW()			Last updated timestamp

Role_permissions

Field Name	Data Type	Constraint	PK	FK	Description
id	INTEGER	PRIMARY KEY	**Y**		Auto-increment ID
role_id	INTEGER	NOT NULL		**Y**	FK → roles.id
permission_id	INTEGER	NOT NULL		**Y**	FK → permissions.id
created_at	TIMESTAMP	DEFAULT NOW()			Assignment timestamp

Refresh_tokens

Field Name	Data Type	Constraint	PK	FK	Description
id	INTEGER	PRIMARY KEY	**Y**		Auto-increment ID
token_id	UUID	UNIQUE NOT NULL			Unique token UUID
expire_at	TIMESTAMP WITH TIME ZONE	NOT NULL			Token expiry datetime
is_revoked	BOOLEAN	DEFAULT FALSE			Whether token has been revoked
created_at	TIMESTAMP	DEFAULT NOW()			Token creation timestamp

Patient_profiles

Field Name	Data Type	Constraint	PK	FK	Description
id	INTEGER	PRIMARY KEY	**Y**		Auto-increment patient profile ID
user_id	INTEGER	UNIQUE NOT NULL		**Y**	FK → users.id (1:1 relationship)
date_of_birth	DATE	NULLABLE			Patient date of birth
gender	VARCHAR(20)	NULLABLE			Patient gender
address	TEXT	NULLABLE			Patient home address
preferred_language	VARCHAR(50)	DEFAULT 'English'			Preferred language for communication
is_active	BOOLEAN	DEFAULT TRUE			Whether patient profile is active
last_login_at	TIMESTAMP	NULLABLE			Last login timestamp
created_at	TIMESTAMP	DEFAULT NOW()			Profile creation timestamp
updated_at	TIMESTAMP	DEFAULT NOW()			Last updated timestamp

Providers

Field Name	Data Type	Constraint	PK	FK	Description
id	INTEGER	PRIMARY KEY	**Y**		Auto-increment provider ID
full_name	VARCHAR(255)	NOT NULL			Doctor/provider full name
specialization	VARCHAR(255)	NOT NULL			Medical specialization e.g. Cardiology
qualification	VARCHAR(255)	NOT NULL			Degrees and qualifications
experience	INTEGER	NULLABLE			Years of experience
phone	VARCHAR(20)	NULLABLE			Provider contact phone
email	VARCHAR(255)	UNIQUE NULLABLE			Provider email address
bio	TEXT	NULLABLE			Provider biography and profile text
is_active	BOOLEAN	DEFAULT TRUE			Whether provider is currently active
created_at	TIMESTAMP	DEFAULT NOW()			Record creation timestamp
updated_at	TIMESTAMP	DEFAULT NOW()			Last updated timestamp

Appointment_types

Field Name	Data Type	Constraint	PK	FK	Description
id	INTEGER	PRIMARY KEY	**Y**		Auto-increment appointment type ID
name	VARCHAR(100)	UNIQUE NOT NULL			Type name e.g. General Consultation
description	TEXT	NULLABLE			Detailed description of appointment type
duration_minutes	INTEGER	NOT NULL			Default duration for this type in minutes
instructions	TEXT	NULLABLE			Pre-visit instructions sent in confirmation
is_active	BOOLEAN	DEFAULT TRUE			Whether this type is currently available
created_at	TIMESTAMP	DEFAULT NOW()			Record creation timestamp

Provider_appointment_types

Field Name	Data Type	Constraint	PK	FK	Description
id	INTEGER	PRIMARY KEY	**Y**		Auto-increment ID
provider_id	INTEGER	NOT NULL		**Y**	FK → providers.id
appointment_type_id	INTEGER	NOT NULL		**Y**	FK → appointment_types.id
is_active	BOOLEAN	DEFAULT TRUE			Whether this provider offers this appointment type
created_at	TIMESTAMP	DEFAULT NOW()			Record creation timestamp

Available_slots

Field Name	Data Type	Constraint	PK	FK	Description
id	INTEGER	PRIMARY KEY	**Y**		Auto-increment slot ID
provider_id	INTEGER	NOT NULL		**Y**	FK → providers.id
availability_date	DATE	NOT NULL			Date of the available slot
start_time	TIME	NOT NULL			Slot start time
end_time	TIME	NOT NULL			Slot end time
status	ENUM('available','booked','blocked','on_hold')	DEFAULT 'available'			Current slot status

Field Name	Data Type	Constraint	PK	FK	Description
created_by	INTEGER	NOT NULL		**Y**	FK → users.id (front desk user who created slot)
notes	TEXT	NULLABLE			Optional notes on the slot
created_at	TIMESTAMP	DEFAULT NOW()			Record creation timestamp
updated_at	TIMESTAMP	DEFAULT NOW()			Last updated timestamp

Appointments

Field Name	Data Type	Constraint	PK	FK	Description
id	INTEGER	PRIMARY KEY	**Y**		Auto-increment appointment ID
patient_id	INTEGER	NOT NULL		**Y**	FK → patient_profiles.id
provider_id	INTEGER	NOT NULL		**Y**	FK → providers.id
appointment_type_id	INTEGER	NOT NULL		**Y**	FK → appointment_types.id
availability_slot_id	INTEGER	UNIQUE NOT NULL		**Y**	FK → available_slots.id (unique to prevent double-booking)
session_id	INTEGER	NULLABLE		**Y**	FK → conversation_sessions.id (booking session)
scheduled_date	DATE	NOT NULL			Date of appointment
scheduled_start_time	TIME	NOT NULL			Appointment start time
scheduled_end_time	TIME	NOT NULL			Appointment end time
status	ENUM('scheduled','confirmed','cancelled','completed','no_show')	DEFAULT 'scheduled'			Appointment lifecycle status
reason_for_visit	TEXT	NULLABLE			Patient-stated reason for visit
notes	TEXT	NULLABLE			Internal notes on the appointment
booking_channel	ENUM('voice','outbound_call','chat','manual')	NOT NULL			Channel through which appointment was booked
instructions	TEXT	NULLABLE			Instructions sent to patient (overrides appointment_type instructions)
cancelled_at	TIMESTAMP	NULLABLE			Timestamp when appointment was cancelled
cancellation_reason	TEXT	NULLABLE			Reason for cancellation

Field Name	Data Type	Constraint	PK	FK	Description
created_at	TIMESTAMP	DEFAULT NOW()			Record creation timestamp
updated_at	TIMESTAMP	DEFAULT NOW()			Last updated timestamp

Conversation_sessions

Field Name	Data Type	Constraint	PK	FK	Description
id	INTEGER	PRIMARY KEY	**Y**		Auto-increment session ID
patient_id	INTEGER	NULLABLE		**Y**	FK → patient_profiles.id (null if unverified at session start)
channel	ENUM('voice','outbound_call','chat')	NOT NULL			Communication channel for session
phone_no	VARCHAR(20)	NULLABLE			Phone number used in this session
status	ENUM('active','completed','dropped','failed')	DEFAULT 'active'			Current session status
intent	TEXT	NULLABLE			Primary detected patient intent
last_flow_step	TEXT	NULLABLE			Last step in conversation flow for recovery
context_snapshot	JSONB	NULLABLE			LangGraph state snapshot for session recovery
started_at	TIMESTAMP	DEFAULT NOW()			Session start timestamp
ended_at	TIMESTAMP	NULLABLE			Session end timestamp
duration_seconds	INTEGER	NULLABLE			Total session duration in seconds
identity_verified	BOOLEAN	DEFAULT FALSE			Whether patient identity was confirmed
booking_completed	BOOLEAN	DEFAULT FALSE			Whether a booking was completed in this session
created_at	TIMESTAMP	DEFAULT NOW()			Record creation timestamp

Conversation_turns

Field Name	Data Type	Constraint	PK	FK	Description
id	INTEGER	PRIMARY KEY	**Y**		Auto-increment turn ID
session_id	INTEGER	NOT NULL		**Y**	FK → conversation_sessions.id

Field Name	Data Type	Constraint	PK	FK	Description
turn_order	INTEGER	NOT NULL			Sequential order of this turn within the session
speaker	ENUM('patient','assistant')	NOT NULL			Who spoke in this turn
message	TEXT	NOT NULL			Transcribed or generated message content
intent_detected	VARCHAR(100)	NULLABLE			NLU intent detected in this turn
entities	JSONB	NULLABLE			Entities extracted: date, time, provider, appointment type
confidence_score	DECIMAL(4,3)	NULLABLE			NLU confidence score (0.000 to 1.000)
created_at	TIMESTAMP	DEFAULT NOW()			Turn creation timestamp

Outbound_call_tasks

Field Name	Data Type	Constraint	PK	FK	Description
id	INTEGER	PRIMARY KEY	**Y**		Auto-increment task ID
patient_id	INTEGER	NOT NULL		**Y**	FK → patient_profiles.id
appointment_id	INTEGER	NULLABLE		**Y**	FK → appointments.id (if linked to existing appointment)
trigger_reason	ENUM('new_booking','reschedule','reminder','follow_up','no_show')	NOT NULL			Why this call was triggered
task_status	ENUM('pending','in_progress','completed','failed','cancelled')	DEFAULT 'pending'			Current task processing status
priority	INTEGER	DEFAULT 5			Call priority (1 = highest, 10 = lowest)
max_retries	INTEGER	DEFAULT 3			Maximum number of call attempts allowed
retry_interval_minutes	INTEGER	DEFAULT 60			Minutes to wait between retry attempts
attempt_count	INTEGER	DEFAULT 0			Current number of attempts made
scheduled_at	TIMESTAMP	NOT NULL			When this call should first be attempted
next_attempt_at	TIMESTAMP	NULLABLE			Scheduled timestamp for next retry
last_attempt_at	TIMESTAMP	NULLABLE			Timestamp of most recent attempt
completed_at	TIMESTAMP	NULLABLE			Timestamp when task was completed
notes	TEXT	NULLABLE			Internal notes on this task

created_by	INTEGER	NOT NULL		**Y**	FK → users.id (who created this task)2
created_at	TIMESTAMP	DEFAULT NOW()			Record creation timestamp
updated_at	TIMESTAMP	DEFAULT NOW()			Last updated timestamp

Call_attempts

Field Name	Data Type	Constraint	PK	FK	Description
id	INTEGER	PRIMARY KEY	**Y**		Auto-increment attempt ID
task_id	INTEGER	NOT NULL		**Y**	FK → outbound_call_tasks.id
session_id	INTEGER	NULLABLE		**Y**	FK → conversation_sessions.id (if call was answered)
attempt_number	INTEGER	NOT NULL			Sequential attempt number for this task
phone_dialed	VARCHAR(20)	NOT NULL			Phone number that was dialed
dialed_at	TIMESTAMP	NOT NULL			When the call was initiated
answered_at	TIMESTAMP	NULLABLE			When the call was answered (null if no answer)
ended_at	TIMESTAMP	NULLABLE			When the call ended
duration_seconds	INTEGER	NULLABLE			Total call duration in seconds
outcome	ENUM('answered','no_answer','voicemail','busy','failed','booked','declined')	NOT NULL			Outcome of this call attempt
disconnect_point	VARCHAR(100)	NULLABLE			Flow step at which call disconnected
voicemail_left	BOOLEAN	DEFAULT FALSE			Whether a voicemail message was left
error_code	VARCHAR(50)	NULLABLE			Telephony error code if call failed
error_message	TEXT	NULLABLE			Detailed error message
created_at	TIMESTAMP	DEFAULT NOW()			Record creation timestamp

Notifications

Field Name	Data Type	Constraint	PK	FK	Description
id	INTEGER	PRIMARY KEY	**Y**		Auto-increment notification ID
appointment_id	INTEGER	NULLABLE		**Y**	FK → appointments.id

Field Name	Data Type	Constraint	PK	FK	Description
patient_id	INTEGER	NOT NULL		**Y**	FK → patient_profiles.id
channel	ENUM('email','sms')	NOT NULL			Delivery channel for notification
type	ENUM('booking_confirmation','reminder','cancellation','reschedule')	NOT NULL			Notification event type
recipient_address	TEXT	NOT NULL			Email address or phone number of recipient
message_body	TEXT	NOT NULL			Rendered notification message content
status	ENUM('pending','sent','delivered','failed')	DEFAULT 'pending'			Delivery status of notification
payload	JSONB	NULLABLE			Full provider API payload and response for debugging
sent_at	TIMESTAMP	NULLABLE			Timestamp when notification was sent
delivered_at	TIMESTAMP	NULLABLE			Timestamp when delivery was confirmed
error_message	TEXT	NULLABLE			Error details if notification failed
created_at	TIMESTAMP	DEFAULT NOW()			Record creation timestamp

ER Diagram

Relationships

- users (1) → (1) patient_profiles via user_id
- users (N) → (1) roles via role_id
- roles (M) ↔ (N) permissions via role_permissions
- patient_profiles (1) → (N) appointments via patient_id
- providers (1) → (N) available_slots via provider_id
- providers (M) ↔ (N) appointment_types via provider_appointment_types
- appointment_types (1) → (N) appointments via appointment_type_id
- available_slots (1) → (1) appointments via availability_slot_id (unique constraint prevents double-booking)
- conversation_sessions (1) → (N) conversation_turns via session_id
- conversation_sessions (1) → (0..1) appointments via session_id
- patient_profiles (1) → (N) outbound_call_tasks via patient_id
- outbound_call_tasks (1) → (N) call_attempts via task_id
- call_attempts (1) → (0..1) conversation_sessions via session_id
- appointments (1) → (N) notifications via appointment_id

Failure Handling

Failure Scenario	Detection	Recovery Action
API timeout	Request timeout > 30s	Retry with exponential backoff, max 3 attempts
Database connection lost	Connection pool exhausted	Circuit breaker activates; failover to read replica
Voice call drops mid-conversation	Telephony disconnect event	Session marked 'dropped'; context_snapshot preserved for re-initiation
Outbound call not answered	Outcome = no_answer or busy	next_attempt_at scheduled per retry_interval_minutes; max_retries enforced before marking failed
LangGraph worker crash	Worker heartbeat failure in Redis	Kubernetes restart policy triggers; session recoverable from context_snapshot
SMS/Email delivery failure	Provider returns non-2xx	Notification marked 'failed'; alert triggered; record queued for manual follow-up
Slot double-booking race condition	Unique constraint violation on availability_slot_id	Transaction rolled back; assistant notifies patient and offers next available slot
Invalid patient identity	Confidence below threshold after max attempts	Call escalated to front desk or session ended gracefully with instructions
NLU intent confidence low	confidence_score below 0.60	Assistant asks clarifying question; turn logged with low confidence flag for review